

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Concept Mapping (Medium)

Assessment Tool Weightage: 35%

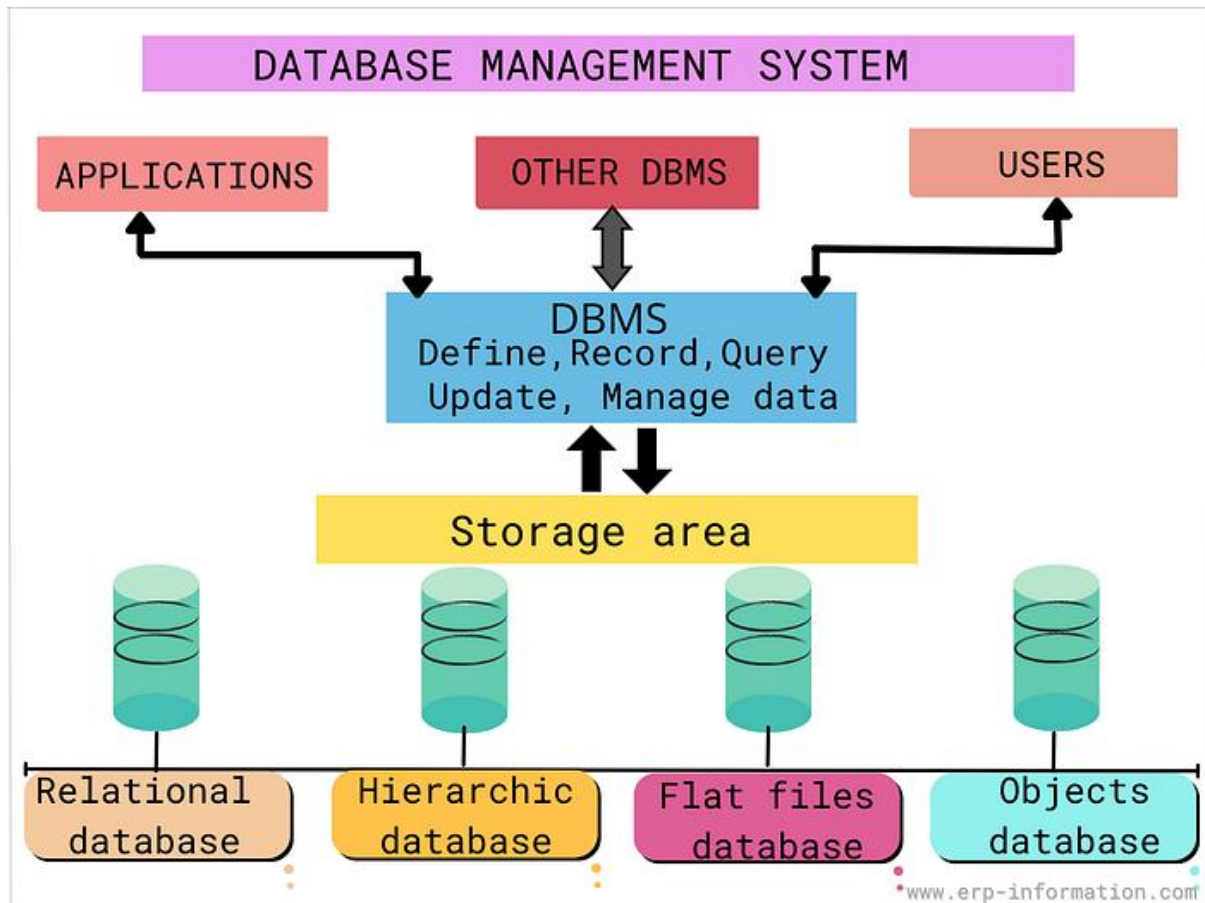
Code of Conduct:

I __K.Sandhya (192424061)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

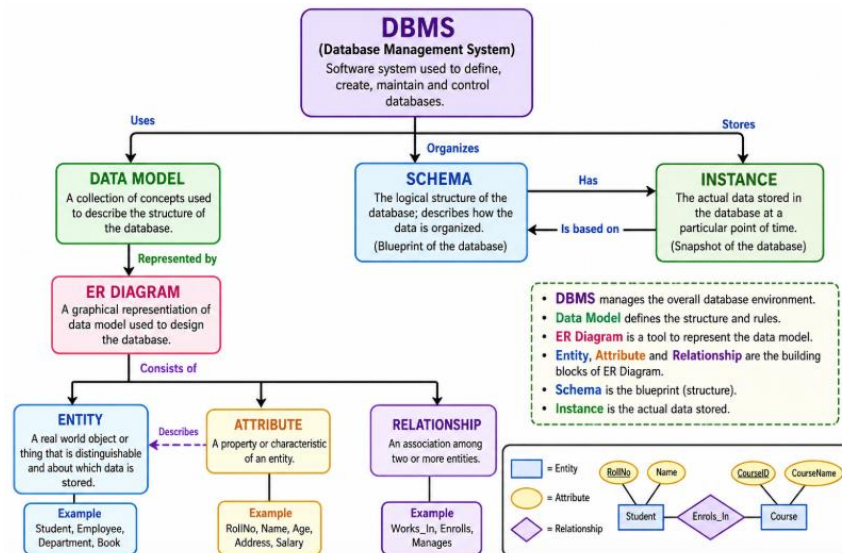
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand



QUESTIONS:

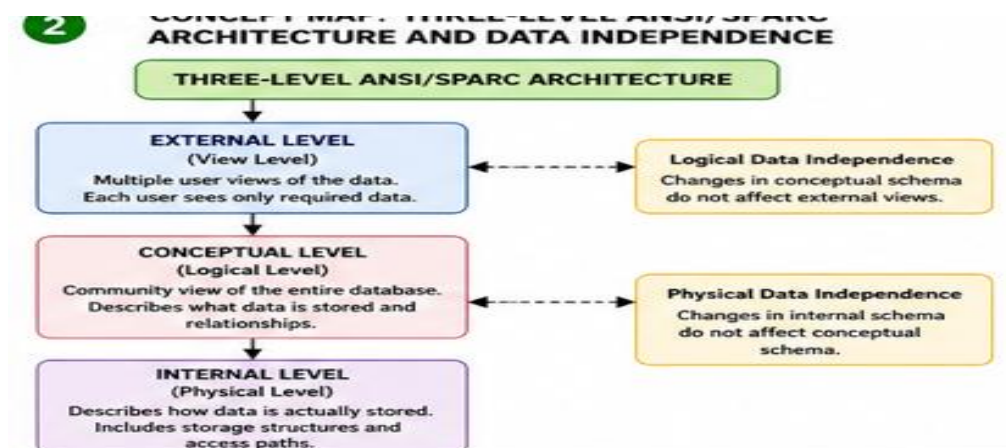
1. Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.

ANSWER:



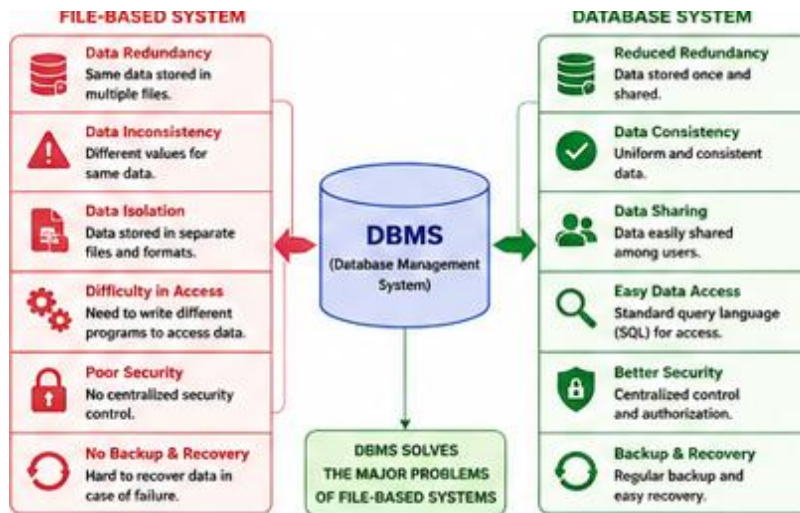
2. Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.

ANSWER:



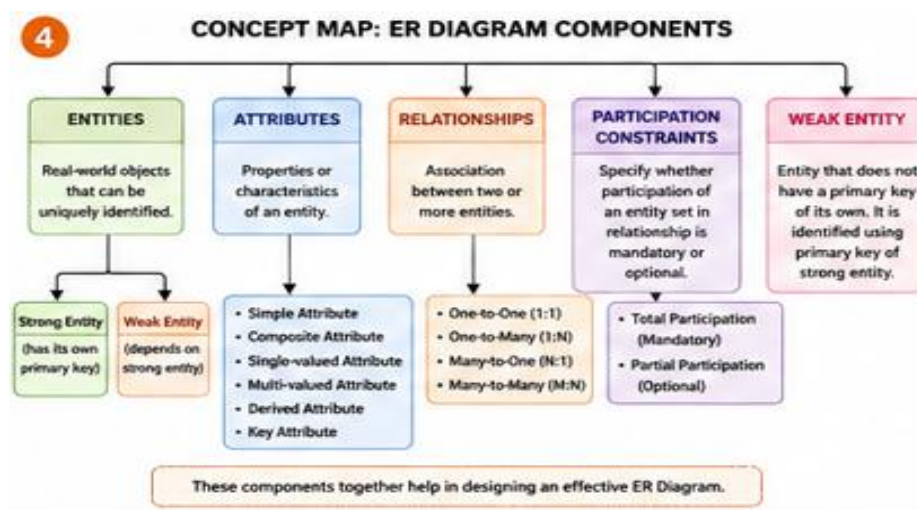
3. Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.

ANSWER :



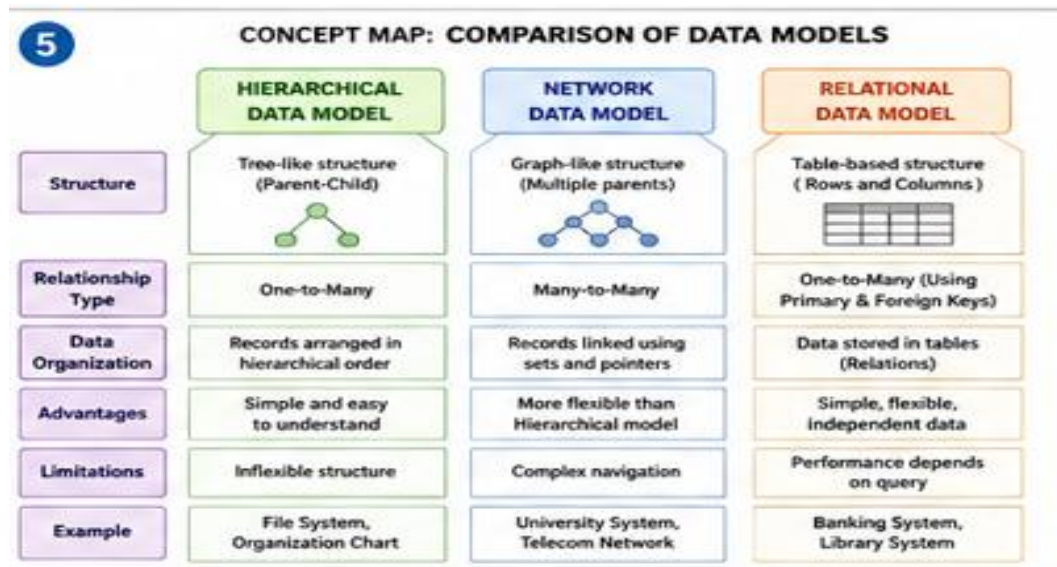
4. Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.

ANSWER :



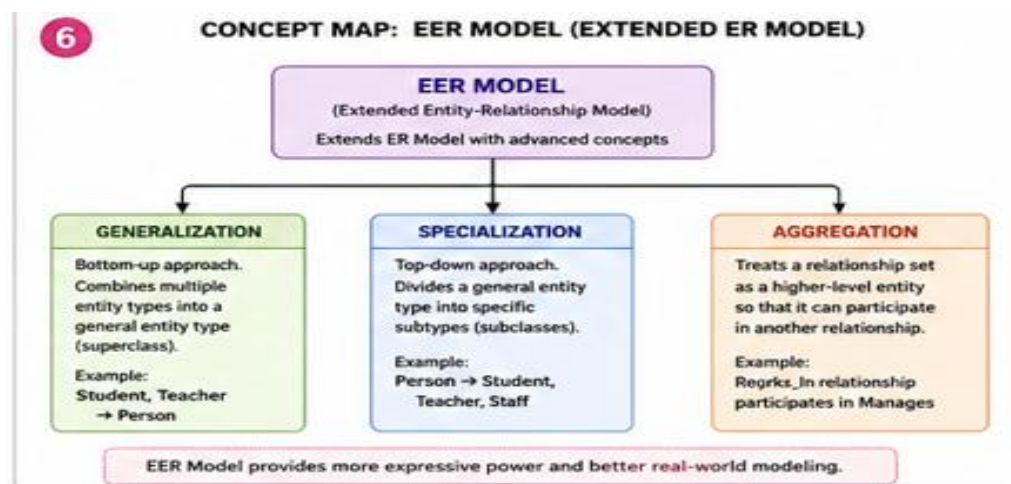
5. Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.

ANSWER :



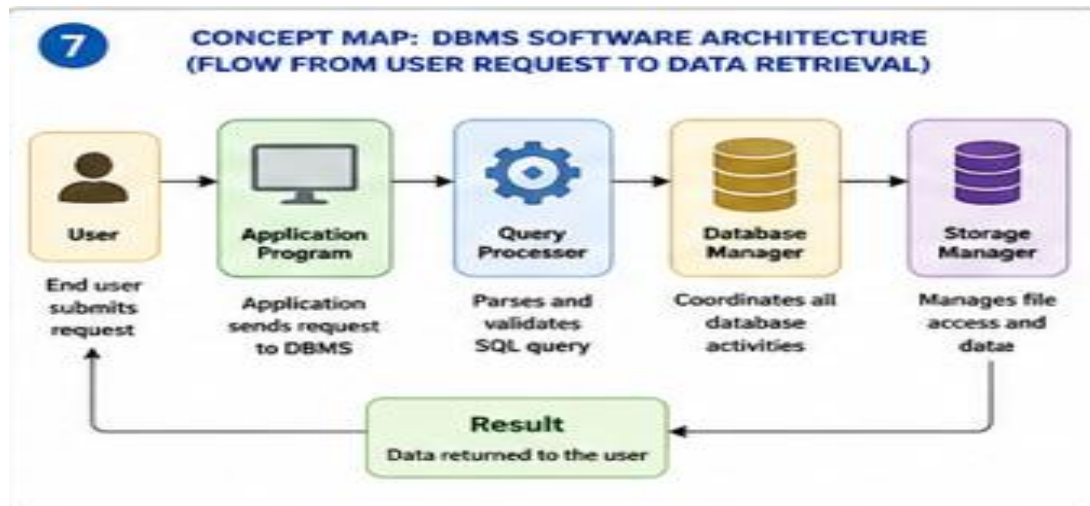
6. Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.

ANSWER :



7. Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.

ANSWER :



8. Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.

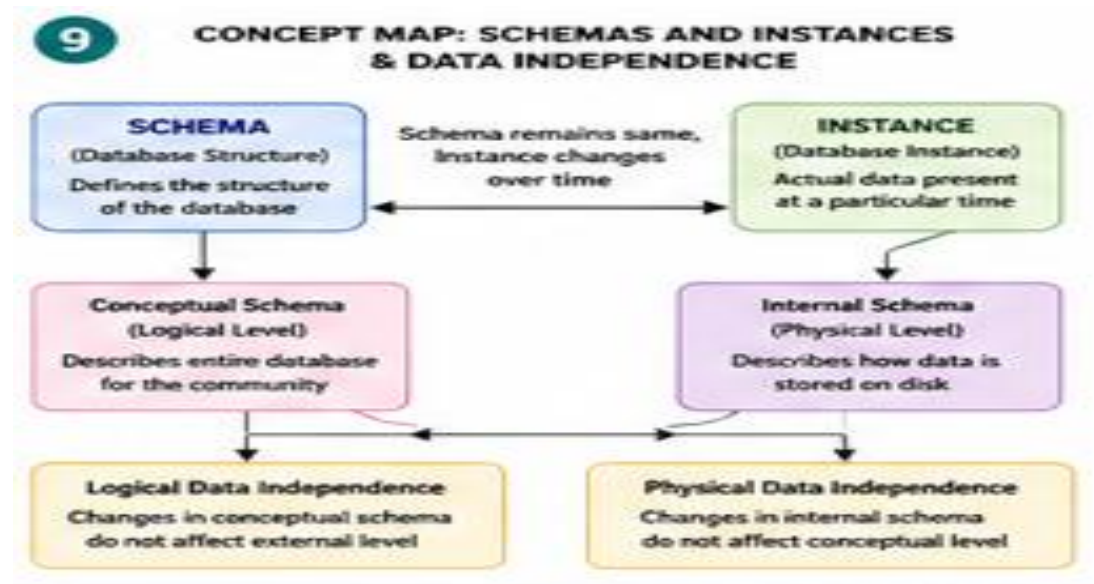
ANSWER :

8 CONCEPT MAP: DATA MODELS AND REAL-WORLD APPLICATIONS

DATA MODEL	SUITABLE FOR	EXAMPLE APPLICATIONS
Hierarchical Model	Data with parent-child relationship and strict hierarchy	- File Systems - Organization Structure - XML Data
Network Model	Complex many-to-many relationships and high-performance needs	- Airline Reservation System - Telecom Network - Banking System
Relational Model	General-purpose applications requiring simplicity and flexibility	- College Management System - Hospital Management System - E-commerce, HR System

9. Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.

ANSWER :



10 Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.

ANSWER :

